NUQA

A Cleaner City, Through Intelligence

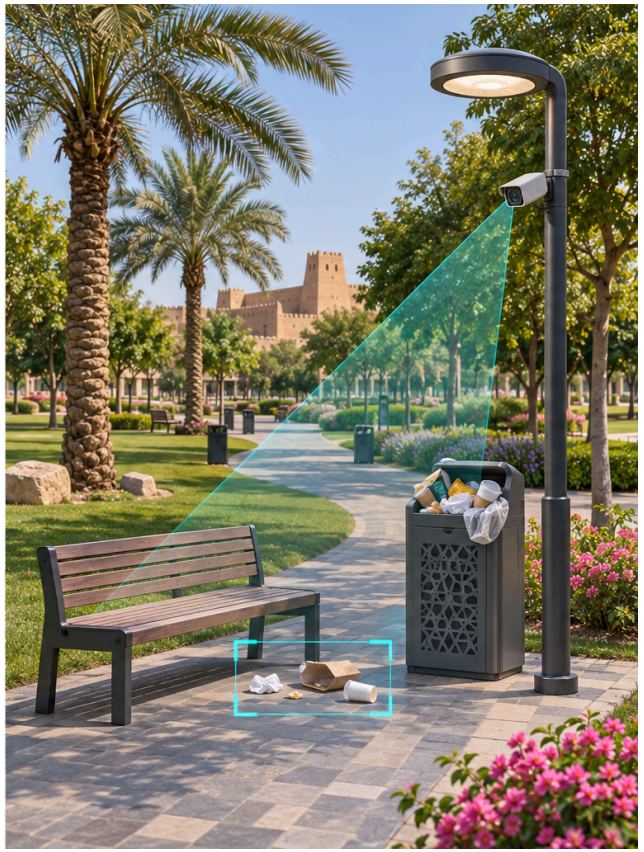
Connect observation, response and verified action.

SMART CAMERAS + CONNECTED PEOPLE



An ordinary afternoon. A problem that keeps growing.

Illustrative scenario: one overflowing park bin affects visitors, the environment and the cleaning team.



01 A family arrives

The nearest bin is full. Waste has collected beside the seating area.

02 The impact spreads

Visitors avoid the area; loose litter can reach planted ground or drains.

03 The team lacks a clear signal

Without a timely, located alert, staff must find and prioritize the cleanup.

The unmet need: detect it, locate it and give someone clear responsibility.

The scale behind the scene: global plastic-waste data

OECD data for 2019, published in 2022. Percentages refer to global plastic waste by treatment route.

22%

Mismanaged globally

9%

Recycled globally

Why it matters

Uncontrolled waste can reach soil and waterways, harm wildlife and reduce the quality of shared places.

Global plastic-waste treatment mix



Source: OECD, Global Plastics Outlook (2022), data year 2019. doi.org/10.1787/de747aef-en

The problem NUQA is designed to address

Litter can harm the environment. NUQA connects detection to a coordinated response.

01

Environmental harm

Litter can contaminate soil and water, harm wildlife and degrade shared green spaces.

02

Littering lacks context

A sighting alone may not reliably establish when, where or from which vehicle waste was thrown.

03

Response is fragmented

Disconnected reports, assignments and updates make ownership and closure harder to follow.

The solution: one connected cleanliness platform

NUQA brings smart-camera alerts, citizen reports and field work into a shared case record.

01

Observe

Camera signals + citizen reports
Location, time and event context.

02

Coordinate

Staff review + task assignment
A clear owner and next action.

03

Resolve

Field response + verification
Citizens receive status updates.

The intended result: earlier awareness, clearer responsibility and traceable resolution.

Artificial intelligence analyzes the scene

Proposed edge-AI processing turns camera frames into a structured event for review.

01

Video frames

Capture the approved road or public-space zone.

02

Detect and track

Locate waste, bins and relevant object movement.

03

Classify the event

Flag overflow, scattered litter or a potential throwing event.

04

Create an alert

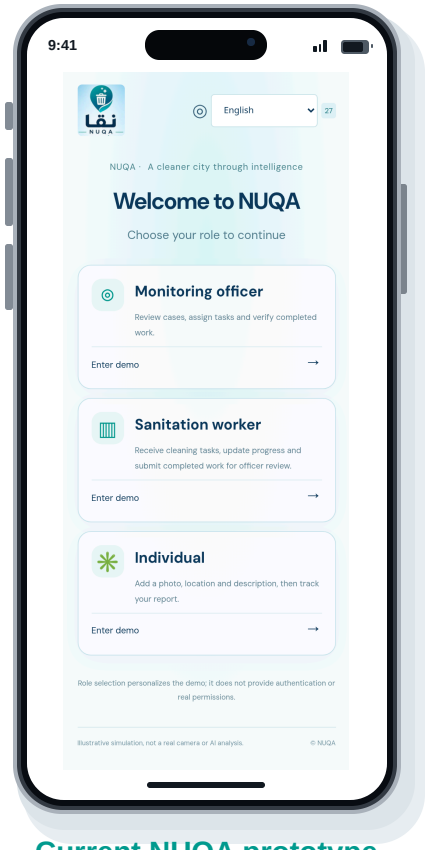
Attach event type, time, location and selected evidence.

Roads: link a potential throwing event to the vehicle; read a visible plate when permitted.

Parks: analyze waste around bins, benches and paths, then route the event for staff review.

Three roles. A focused experience for each.

The working website starts with role selection and offers 27 interface languages.



Current NUQA prototype

01 Individual

Report a location and issue; follow its status.

02 Sanitation worker

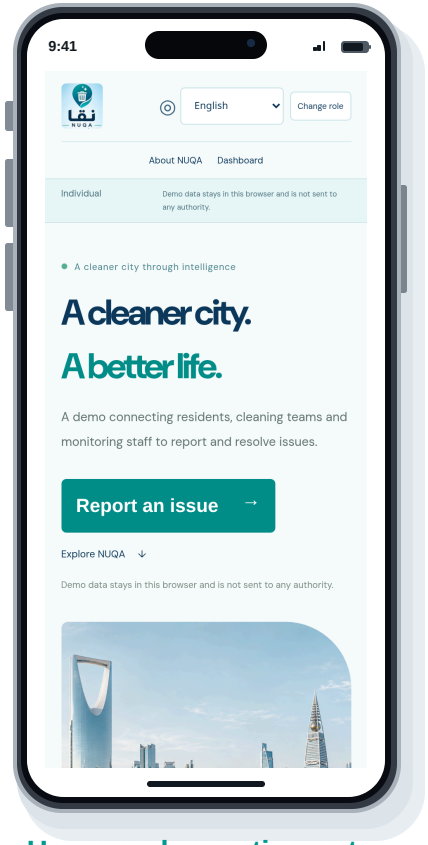
Start assigned work and submit completion for review.

03 Monitoring officer

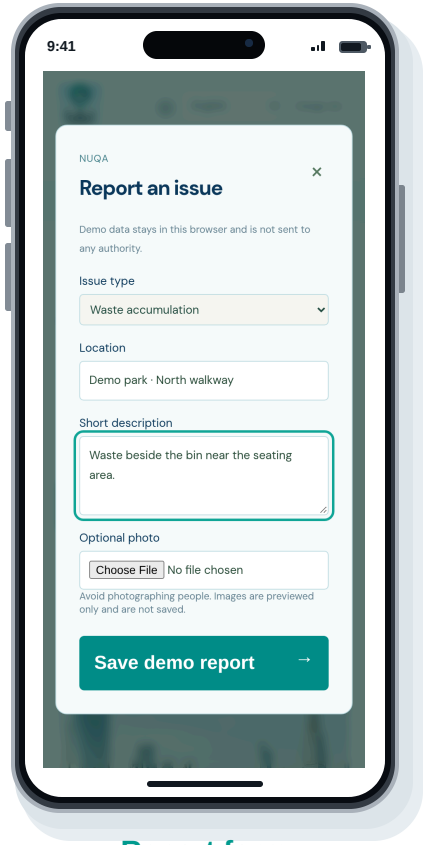
Review camera events, manage cases and verify closure.

Citizen reporting: simple inputs, visible progress

A short report connects a local observation to the same operational case record.



Home and reporting entry



Report form

01 Describe the issue

Choose the issue type, add a location and write a short description.

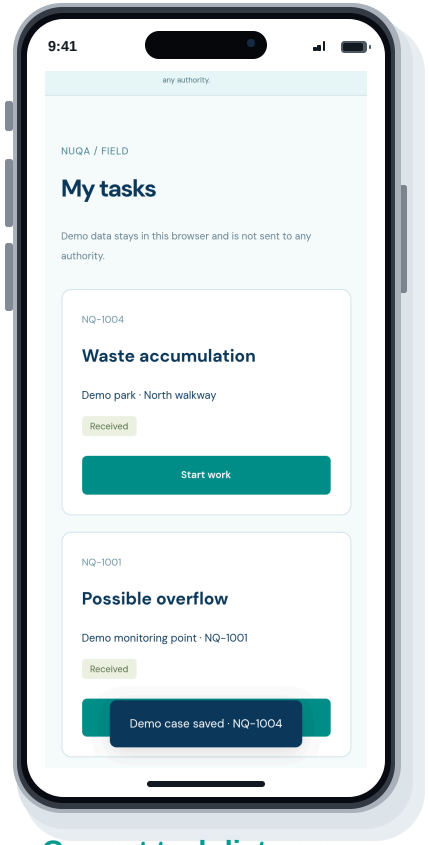
02 Preview, then submit

An optional photo adds context. Status updates keep the report trackable.

Current demo: reports stay in the browser; photos are previewed, not stored.

Sanitation teams: turn cases into accountable work

Tasks move forward with a defined completion handoff to monitoring staff.



Current task-list screen

01 Receive and start

A worker sees a task, issue type, location and current status.

02 Confirm completion

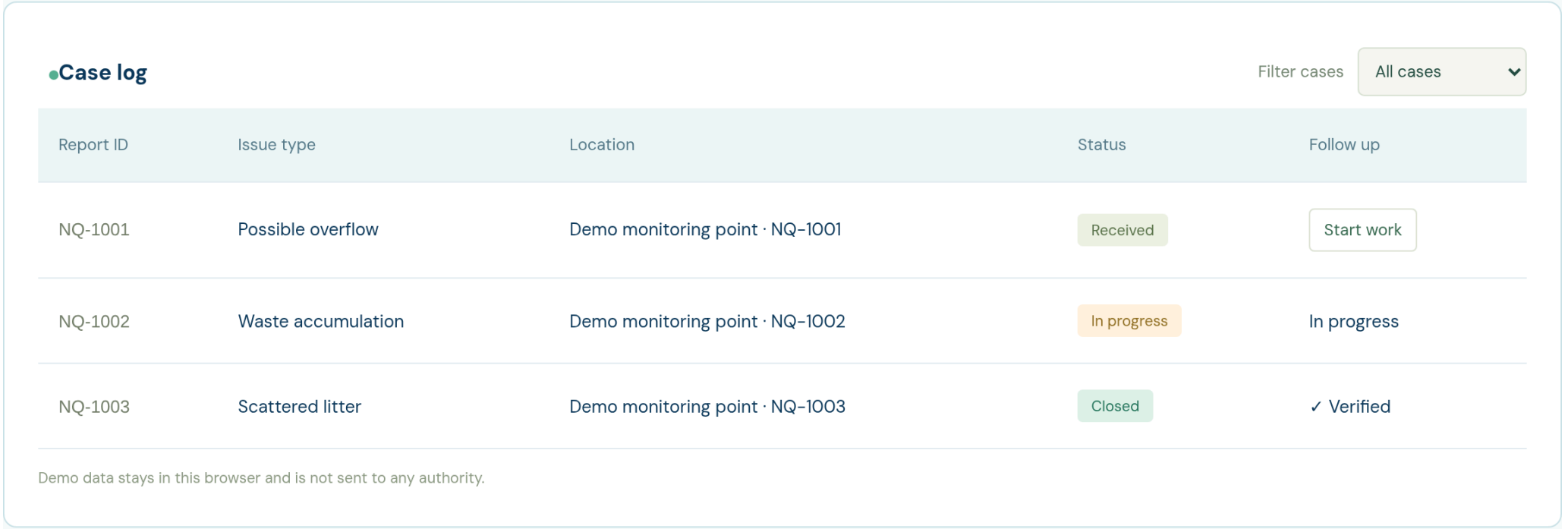
After cleanup, the worker submits the task for staff review.

03 Verify before closure

The officer checks the outcome and closes the case.

Municipal view: one record from report to closure

Staff can inspect status, initiate work and verify completed tasks.



Case log

Filter cases: All cases

Report ID	Issue type	Location	Status	Follow up
NQ-1001	Possible overflow	Demo monitoring point · NQ-1001	Received	Start work
NQ-1002	Waste accumulation	Demo monitoring point · NQ-1002	In progress	In progress
NQ-1003	Scattered litter	Demo monitoring point · NQ-1003	Closed	✓ Verified

Demo data stays in this browser and is not sent to any authority.

Unified cases

Camera events and citizen reports

Reviewed transitions

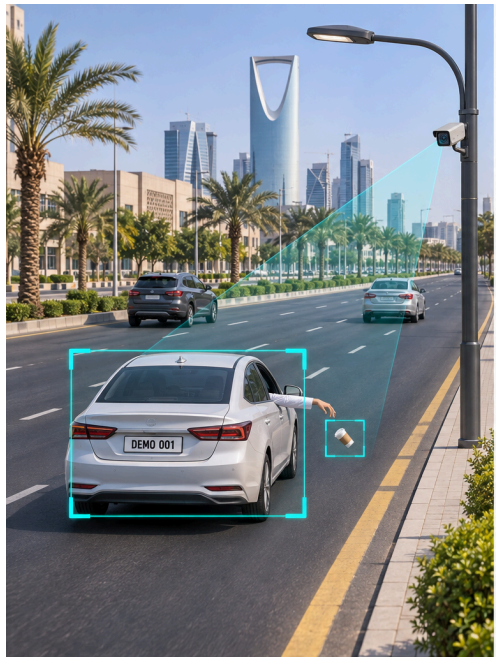
Received → In progress → Review

Verified closure

Officer confirmation before close

Roadside monitoring: event, vehicle and plate

Proposed AI captures the incident context; the website demo uses fictional vehicles and plates.



Camera and event concept

Illustrative scene



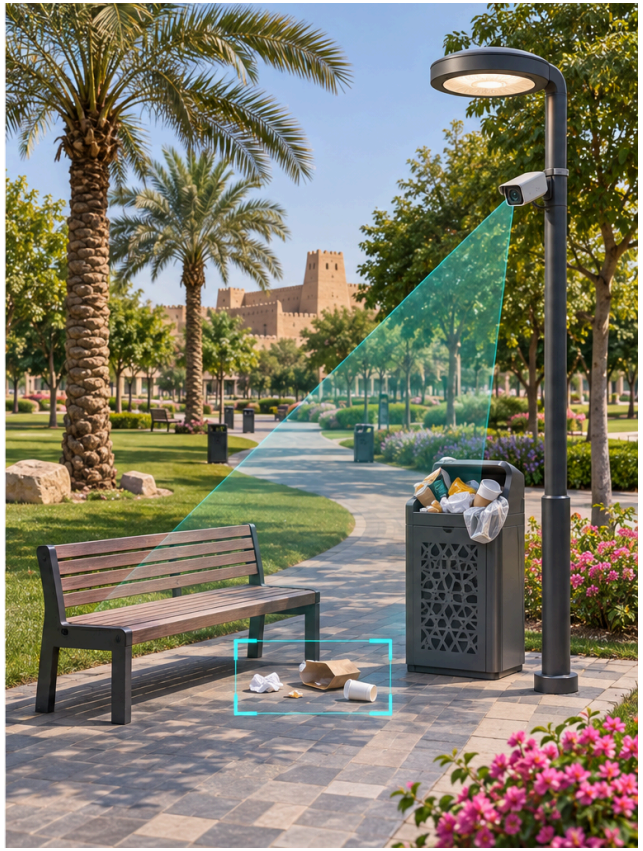
Current vehicle simulation

FICTIONAL PLATE: DEMO-001

Staff review before real action

Park simulation: AI flags litter near a seating area

Lighting-pole camera → AI analysis → Staff review → Cleaning response.



Illustrative scene and simulated event

PARK-DEMO-01 / LP-CAM-01

AI EVENT SIMULATION

Detected condition

Scattered litter beside a bench



LITTER-01

Event context

Camera zone + time + selected image

AI flag → Staff review → Cleaning task

Designed for the places people share

Concept settings for an authorized pilot, using cameras on existing lighting poles.



Public parks

Bins, lawns and seating areas



Walkways

Paths and shared pedestrian zones



Public plazas

Gathering areas and planted edges

The team

NUQA — A cleaner city through intelligence.

Manar Al-Nuqaydan

Tamadher

The next step

Pilot next: measure detection quality, response time and verified cleanup.